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Original Article

Effect of Concussion on Performance of National Football League Players

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ABSTRACT:

Purpose: Lingering neurologic injury after concussion may expose athletes to increased risk if return-to-play is premature. We explored whether on-field performance after concussion is a marker of lingering neurologic injury. **Methods:** Retrospective cohort study on 1,882 skill-position players who played in the National Football League (NFL) during 2007-2010. Players with concussion based on the weekly injury report, were compared to players with other head and neck injuries (controls) on measures of on-field performance using Football Outsiders' calculation of Defense-Adjusted Yards Above Replacement (DYAR): a measure of a player's contribution controlling for game context. Changes in performance, relative to a player's baseline level of performance, were estimated before and after injury using fixed-effects models. **Results:** The study included 140 concussed players and 57 controls. Players with concussion performed no better or worse than their baseline upon return-to-play. However, a decline in DYAR, relative to their prior performance, was noted two weeks and one week prior to appearing on the injury report. Concussed players performed slightly better than controls in situations where they returned to play the same week as appearing on the injury report. **Conclusions:** Upon return, concussed NFL players performed at their baseline level of performance, suggesting that players have recovered from concussion. Decline in performance noted 2 weeks and 1 week prior to appearing on the injury report may suggest that concussion diagnosis was delayed or that concussion can be a multi-hit phenomenon. Athletic performance may be a novel tool for assessing concussion injury and recovery.

Key words: Concussion, NFL, American football, Head Injury, Performance

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INTRODUCTION:

Published consensus statements recommend that concussed athletes return to play only after neurologic symptoms, cognitive function, and balance have returned to baseline.[1-4] However, it is unknown whether current clinical evaluations are sufficiently sensitive to ensure that on-going brain dysfunction has resolved. Few studies have sought to examine athletic performance as a measure of neurologic health and concussion recovery. Only one study to our knowledge, has looked at the short-term effect of concussion on performance after return to play in football players.[5]

Athletic performance following concussion may be a novel measure of on-going brain dysfunction. While a decline in performance does not necessarily imply on-going brain dysfunction , it is plausible that they are linked . Brain networks involved in saccadic eye movements and eye tracking, convergence, attention, balance, coordination, processing speed, and reaction time are known to be effected in acute concussion. [6-11] Neurocognitive testing has become a mainstay for concussion assessment but alone, it may fall short of assessing the complex network dysfunction of concussion, and may miss lingering neurologic dysfunction even after symptom resolution. (Athletic tasks, such as running a route to catch a football, are complex and involve visuospatial processing, rapid decision-making, reaction time, and complex coordination. More complex cognitive tasks require more extensive brain networks, and in turn, may be more sensitive to changes from mild traumatic brain injury.[12-16]

We sought to explore the impact of concussion on overall on-field performance prior and subsequent to a reported concussion on the National Football League (NFL) injury report in NFL football players using advanced statistical performance data from Football Outsiders.[17] If subtle brain dysfunction persists after concussion, despite presumed clinical symptom recovery, we

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hypothesized that players would demonstrate a decline in performance after returning to the field compared to their prior performance and to players with other injuries.

METHODS:

We performed a retrospective cohort study of NFL players who sustained concussion, examining player performance to estimate the impact of reported concussion on overall performance. Our primary analysis estimated the impact of reported concussion on performance comparing a player to their baseline level of performance. Secondary analysis compared concussed players to controls to explore if the temporal relationship between concussion and performance was similar to other injuries. The study was approved by the institutional review board (IRBMED HUM00097511).

Study Population and Exposure

Our population consisted of all NFL "skill position players" (running backs, quarterbacks, wide receivers and tight ends) who had at least one touch in live regular season play during 2007-2010. This is an entirely male population. The primary exposure of interest was concussion, as reported by the weekly NFL injury report. The NFL injury report is a report compiled by the training and coaching staff each week of the team's injured players. This list must be submitted to league headquarters the Friday before each game and should be updated on the weekend as needed. This list reflects the new injuries of the prior game and practice week as well as persistent injuries with the purpose of alerting league administration and other teams of athletes that may not play in that week's upcoming competition. While teams may not fully report injuries, the league office can fine teams for inaccurate injury reporting, although this is infrequently done.[18] In this study, any injury with "concussion" or "head injury" on the injury report description was classified as a concussion. Control players were players that sustained

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other injuries of the head and neck reported via the weekly NFL injury report and compiled by Football Outsiders.[17] Players that appeared on the injury report with multiple injuries were excluded.

Outcome

Football Outsiders has developed advanced statistics to measure on-field performance, termed "Defense-Adjusted Yards Above Replacement (DYAR)".[17] Similar statistical models have been used in the past.[5] Every single play in the NFL gets a "success value" based on this system, and then that number gets compared to the average success values of plays in similar situations for all players, adjusted for down and distance, field location, time remaining in game, the team's lead or deficit in the game score, and the opposing defense. DYAR estimates an individual skill-position player's aggregate contribution to a team's success based on their combined success value. As a general rule, one above average play is worth about 4-5 points of DYAR. The benefits of DYAR as a statistic is that it compares each play only to plays with similar circumstances, for example, it is more accurate than the conventional NFL conversion statistic because it takes into account that converting third-and-long is more difficult than converting third-and-short, and that a turnover is worse than an incomplete pass because it eliminates the opportunity to move the other team back with a punt on fourth down. It also adjusts for game situation and the quality of the opponent. Another virtue of DYAR is that players at different positions can be compared using the same measurement system and it takes into account the number of plays a player contributes to a given game. The negative of available rate statistics such as Football Outsiders' Defense-adjusted Value Over Average (DVOA), is that they are highly influenced by a small number of plays for players that have a small number of

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touches. Moreover, the number and variety of plays that a player can contribute to may also be a reflection of overall performance that would be discounted with a rate statistic.

Statistical Analysis

To determine the association between concussion and DYAR we developed a multi-level linear regression model predicting weekly DYAR (dependent variable) while allowing for a fixed player-level intercept. Our primary analysis using this approach enables measurement of the effect of concussion on a player's weekly performance compared to their average performance, adjusted for season and week in the season. Regular season games were analyzed only with preseason, postseason and practices being excluded from analysis. Bye weeks were also excluded. To measure the association between concussion and DYAR, we classified each player-week into four broad categories relative to timing of concussion and control injuries: 1) not within four weeks of injury, 2) four or fewer weeks prior to reporting on the injury list, with each individual week analyzed separately 3) during the period the player appeared on the injury report (e.g. a player is injured in week 14, is listed as probable and then plays in week 15), 4) the four weeks after a player appeared on the injury report, each analyzed separately. For each individual week within 4 weeks of the injury period, dummy variables were included in the model to measure the specific time-injury associations. (i.e. week -4, week-3…week +3, week +4) In this framework, time 0 refers to players who played in the same week that they appeared on the injury report (e.g. reported as "probable" due to concussion for a given week, and then played that week). For players that were injured during a game, they would most likely be injured during week -1, the week prior to their appearance on the injury report. Associations between concussion and DYAR were reported with regression coefficients (adjusted difference between observed and expected DYAR) including 95% confidence intervals. Comparisons

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between regression coefficients for concussions and controls were made using Wald tests. All analyses were performed in Stata (StataCorp. 2011. *Stata Statistical Software: Release 12*. College Station, TX: StataCorp LP). Sensitivity analyses were performed that determined whether the primary effects were altered by adjusting for the number of plays per game, the week of the NFL season, player age, number of years in the league, or the number of weeks missed due to injury.

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RESULTS:

We analyzed data on 1,882 unique players over a mean of 1.82 (SD 1.1) seasons. The sample consisted of 982 wide receivers or tight ends (52.2%), 659 running backs (35.0%), and 241 quarterbacks (12.8%), distributions were similar for the population with injuries. Data was available for a mean of 7.4 (SD 9.6) plays/game per player and the primary outcome variable, DYAR was relatively normally distributed with a mean of 6.1 (SD 32.0). There were no differences in the number of games missed for a given injury between concussed athletes (mean 1.2, SD 2.7; median 0, IQR 0-1) and controls (mean 2.0, SD3.7; median 0, IQR 0-3).

A total of 140 concussions were reported on the injury list amongst these players from 2007-2010 and a total of 57 control injuries were detected. Common control injuries were "Neck/Sore neck/Strained Neck" (N=29), "Migraines" (N=4) and "Stinger" (N=4). Most other control injuries appeared to be musculoskeletal injuries to the neck.

Association of Concussion and Performance

Players that played later the same week that they had a concussion disclosed on the injury report had a DYAR that was not significantly different than expected (4.2 95% CI [-2.6, 10.9]). The same was true for players the week after appearing on the injury report for concussion (-2.9 [-9.7, 3.8]). (**Figure 1)** Further, performance after returning to play 2 weeks or later after a

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reported concussion was, on average, no worse than that expected based on prior performance or to controls. In contrast, players had a significantly lower DYAR both two weeks (-7.12 [95% CI -13.0, -1.31]) and one week (-12.2 [-17.9, -6.5]) prior to having a concussion disclosed on the injury report (**Figure 1)**. When compared to controls, those with reported concussions performed worse the week prior to the report (DYAR difference -24.6, $p < 0.01$), but had a non-significantly higher DYAR in the week appearing on the injury report (DYAR difference 11.2, $p = 0.06$). These associations were not substantially altered by adjusting for the number of plays a player played in a given week, the week of the NFL season, or the number of weeks missed due to injury (**Figure 2)**.

DISCUSSION:

We found that NFL players who had a concussion disclosed on an official injury report had no evidence of decline in performance once they were cleared to return. This suggests that NFL players, on the whole, appear to be returning to play after they've sufficiently recovered from concussion and is consistent with prior data on performance after return to play. Kumar et al. looked at NFL players returning from concussion and compared athletic performance pre- and post- injury in those that returned to play within 7 days as well as in those who missed at least 1. They found no difference in performance pre and post injury in either group.[5] Kuhn et al. examined performance in National Hockey League players returning from concussion as compared to controls returning from other leaves and found no difference in performance statistics.[19] Makdissi et al studied performance in Australian rules football players before and after concussion and, similarly, found no change in performance on return to play.[20] However, our study results are in contrast to Wasserman et al, who looked at batting performance in Major League Baseball (MLB) players after concussion and found a decline in nearly all batting metrics

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when comparing pre- and post- concussion and also when compared to control athletes. Despite resolution of symptoms and ImPACT scores, MLB athletes continued to exhibit impaired function in athletic performance.[21]

Interestingly, we also found that players who had a reported concussion had a significantly lower performance in the two weeks prior to appearing on the official injury report. If players' lower performance was isolated to just the week prior to being placed on the NFL injury report, this finding could be explained by the concussive injury occurring that game. However, the finding that players had a lower performance two weeks prior to being listed on the injury report suggests that neuro-injury may have been present prior to the game in which the diagnosed concussion occurred. The two most intuitive, and non-exclusive, interpretations of this finding are: 1. Concussion underreporting or delayed diagnosis and 2. Concussion may be a multi-hit phenomenon. The first interpretation is supported by a 2012 poll of NFL players which showed that 56% of players would hide symptoms of concussion to stay on the field.[22] With continued exertional challenges and lack of adequate physical rest, symptoms can worsen, and may eventually reach a threshold where the injury becomes obvious to the player and observers. The second interpretation raises the possibility that prior diminished performance is due to subclinical injury. This multi-hit hypothesis proposes that there is a cumulative injury process resulting from numerous head impacts per practice and per game. The numerous subconcussive impacts football players are subject to[23] may have subtle deleterious effects including altered functional connectivity,[24] changes in visual working memory, and altered activation of the dorsolateral prefrontal cortex.[25] Subconcussive injury may thus impact elite athletic performance and place the player at increased risk that a subsequent hit will occur, and that it will pass the symptom threshold and present with overt concussive symptoms. This is supported by several

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experimental models which have identified a period of metabolic vulnerability after an initial hit.[26-28] A final explanation to consider is that reduced performance two weeks prior to being listed on the injury report may be a reflection of player fatigue or musculoskeletal injury that then increased a player's risk for subsequent concussion that week or the following week.

PRACTICAL APPLICATIONS:

Coaches and elite athletes need to understand that data from this retrospective cohort study, suggest that there is no evidence of decline in performance once an athlete is cleared to return to play by .

Limitations of this study include the possibility that performance is not a sensitive measure for injury and any lingering effects of concussion simply did not affect performance. The return of DYAR to baseline may then be simply a reflection of performance and not necessarily an indication or confirmation that the concussion has resolved. Secondly, DYAR may not be the optimal measure of individual athletic performance. For example, DYAR cannot assess an individual player's performance without taking into account other players or the entire team's effort on that play. A suboptimal performance measure may lead to the failure to observe an association when one truly exists. Thirdly, there are significant inherent limitations with the NFL injury report, chiefly the diagnoses can be non-specific. Given that our selection of the control group was athletes with other head and neck injuries, this nonspecificity may have falsely identified control group athletes that instead had concussion and the nature of available data made confirming diagnoses via alternate sources infeasible. Concussion under-reporting continues to be a prevalent problem at the NFL level.[22,29] It is possible that some players with concussion were not captured, possibly for strategic reasons on the part of teams, and hence, were assigned to the control group or omitted from the study completely. However, to the extent

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this was the case it would most likely bias our findings to the null. Thus the fact that significant differences in performance within concussed players were detected suggests that this limitation is unlikely to bias our primary conclusions.

CONCLUSION:

In conclusion, we did not find evidence, based on game day performance, that NFL athletes failed to return to baseline prior to return to play after concussion injury during 2007-2010, even prior to the recent era of increased concussion awareness. However, our data also illuminates the potential need for earlier recognition and diagnosis of concussion.

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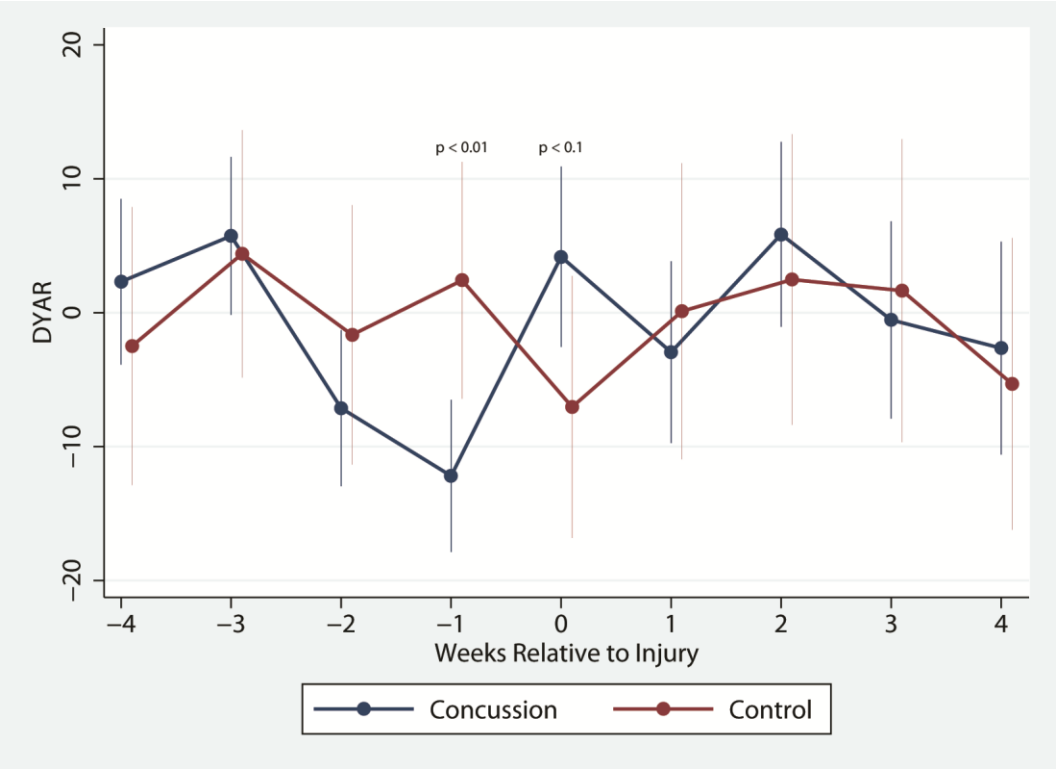

Figure 1: Association between time-to-concussion and performance. The blue lines represent the performance of concussed players relative to their baseline performance in the four weeks prior to appearing on the injury report for concussion (-4 to -1), in weeks where they appeared on the injury report, but played (0) and in the weeks following their concussion (1 to 4). The red lines represent the same associations for controls. Vertical lines represent the 95% confidence intervals of each estimate. Listed p-values compare the differences between performance in concussed and control players.

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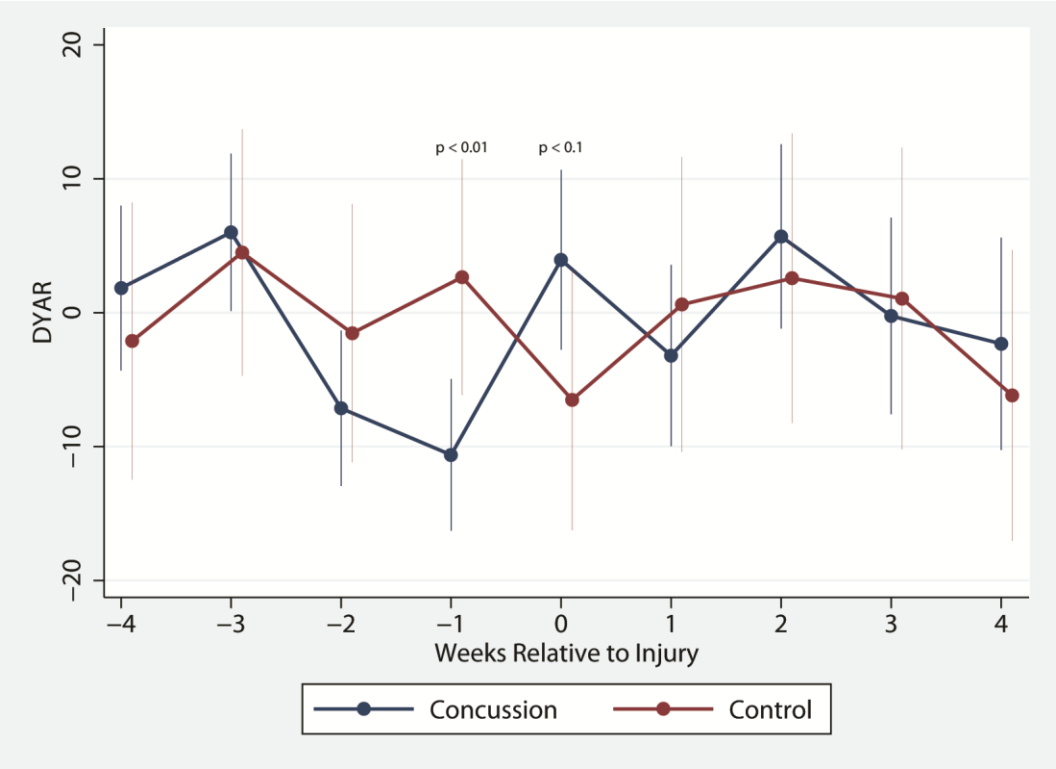

Figure 2: Association between time-to-concussion and performance adjusting for number of plays. The blue lines represent the performance of concussed players relative to their baseline performance in the four weeks prior to appearing on the injury report for concussion (-4 to -1), in weeks where they appeared on the injury report, but played (0) and in the weeks following their concussion (1 to 4). The red lines represent the same associations for controls. Vertical lines represent the 95% confidence intervals of each estimate. Listed p-values compare the differences between performance in concussed and control players.

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